

Exam Date & Time: 11-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - Make up Examination - Jan -2025

Deep learning in Computer Vision [CSE-5041]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1)
 - A) A deep learning engineer is designing a convolutional neural network to detect objects in grayscale images. Each input image is 32×32 pixels in size. One of the convolutional layers in the network applies a series of filters to capture patterns like edges and textures. Consider a 3×3 filter kernel for edge detection:

$$K = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$
 Calculate the output (feature map) size after applying this 3×3 filter to the 28×28 input image, assuming a stride of 1 and "valid" padding (no zero-padding). (5)
 - B) Design a CNN by taking a 3×3 patch from the input image with pixel values I and kernel K as given:

$$K = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \quad I = \begin{pmatrix} 2 & 3 & 0 \\ 4 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$
 Perform the convolution operation by applying the filter K to the patch I . Compute the output value (activation) produced when K is applied to I . Show your calculations step-by-step. (3)
 - C) Describe how building a scale space can increase computational complexity and how deep learning models manage this. (2)
- 2) Discuss the impact of VLAD on retrieval performance, especially in comparison to traditional bag-of-words approaches. Analyze the advantages VLAD offers for handling complex visual patterns in images. (5)
 - A)
 - B) Suppose a sentiment analysis model using BoW is failing to accurately classify certain phrases. How might you modify the feature extraction process to improve context understanding? (3)
 - C) In a dataset with a high percentage of outliers, how does increasing the number of RANSAC iterations impact the model's ability to find a reliable fit, and what trade-offs might arise with this approach? (2)
- 3) How could switching activation functions in some layers help mitigate the vanishing (5)

gradient problem? Suggest specific activation functions that might be beneficial and explain their impact on gradient flow.

- A)
- B) How did ZFNet improve upon AlexNet's design, and what was the purpose of visualizing feature maps in the early layers? (3)
- C) How does DenseNet reduce the number of parameters compared to other deep architectures? (2)
- 4) In a deep learning model using deconvolution layers (also known as transposed convolutions) for image generation, suppose you notice artifacts in the generated images when the deconvolution kernels are initialized randomly. What effect might this have on the output, and how can the choice of kernel initialization (e.g., using learned filters vs. random initialization) and kernel size impact the model's performance? Additionally, explain the potential role of padding strategies in reducing artifacts. (5)
 - A)
 - B) Which deep learning technique is primarily used in neural style transfer for extracting content and style features? (3)
 - C) Suppose you're using Grad-CAM to visualize the important regions for a specific class prediction in a convolutional neural network. After generating the heatmap, you notice that the highlighted regions appear to focus on the background rather than the object of interest in the image. What might be the underlying causes for this unexpected outcome, and how can you modify your Grad-CAM implementation or the model itself to address this issue? (2)
- 5) Discuss the role of Siamese Networks or Triplet Networks in improving verification tasks. How do these architectures help CNNs learn similarity or dissimilarity between faces for verification purposes? (5)
 - A)
 - B) In the context of action/activity recognition using deep learning, you are tasked with developing a model that can recognize human activities from video footage. The video frames contain various motion patterns such as walking, running, sitting, and standing. Explain how 3D Convolutional Neural Networks (3D-CNNs) and Long Short-Term Memory (LSTM) networks can be utilized to address the spatio-temporal nature of human activities. Discuss the advantages of combining these two approaches for more accurate action recognition. (3)
 - C) What is the main advantage of using Fast RNN over traditional RNNs, and how does it achieve improved computational efficiency during training and inference? (2)

-----End-----